Calculating acceleration

AS Level | 9702 AS Ch.2 | Chapter 2: Accelerated motion | Lesson 2.2 | Sections 2.2

Goal: Use a = (v − u)/t and rearrange it for v, u or t.

01 The big picture

For an object with initial velocity u that reaches final velocity v after a time t, the acceleration is the change in velocity divided by the time.

The same equation, rearranged, gives the final velocity: $v = u + at$. To find the time, t is in the denominator, so swap it with the lone term a in one step (cross-multiplication): $t = (v - u)/a$.

Initial velocity u [paraphrase]
The velocity at the start of the time interval.

Final velocity v [paraphrase]
The velocity at the end of the time interval.

02 Key equations

Acceleration

$$a = \frac{v - u}{t}$$

Final velocity

$$v = u + at$$

Time (swap t and a)

$$t = \frac{v - u}{a}$$

Symbols and units

u	initial velocity	$\mathrm{m\,s^{-1}}$	v	final velocity	$\mathrm{m\,s^{-1}}$	t	time taken	s

03 Sample problems with solutions

Sample 1 · Velocity and time

A car moving at 5.0 m s⁻¹ accelerates uniformly at 2.0 m s⁻². Calculate (a) its velocity after 6.0 s and (b) the time it takes to reach 25 m s⁻¹.

Step 1. (a) Final velocity:

$$v = u + at = 5.0 + 2.0 \times 6.0 = 17\ \mathrm{m\,s^{-1}}$$

Step 2. (b) t is in the denominator: swap t and a.

$$t = \frac{v - u}{a} = \frac{25 - 5.0}{2.0} = 10\ \mathrm{s}$$

Answer: (a) 17 m s⁻¹ (b) 10 s

Unit check: $\frac{\mathrm{m\,s^{-1}}}{\mathrm{m\,s^{-2}}} = \mathrm{s}$ ✓

04 Practice problems

P1 A train slows from 20 m s⁻¹ to 8.0 m s⁻¹ in 6.0 s. Calculate its acceleration.
Answer: −2.0 m s⁻²

P2 A stone falls from rest with $a = 9.81$ m s⁻². Find its velocity after 1.5 s.
Answer: 14.7 m s⁻¹ (15 m s⁻¹)

P3 A plane needs 75 m s⁻¹ to take off and accelerates from rest at 2.5 m s⁻². How long does it take?
Answer: 30 s

P4 A cyclist at 6.0 m s⁻¹ decelerates at 0.80 m s⁻². How long until they stop?
Answer: 7.5 s

05 Other useful information

Common mistakes

✗ **Wrong:** Use $v - u$ the wrong way round.
✓ **Right:** Always final minus initial; a negative answer means slowing down.

✗ **Wrong:** Use $a = +0.80$ m s⁻² for a deceleration.
✓ **Right:** Use $a = -0.80$ m s⁻², or the time comes out negative.

Exam tips

- List u, v, a, t with units before you substitute.

What the syllabus asks

- 9702 topic 2: define and use acceleration. Coursebook 2.2.

Explore: Practice questions for this lesson